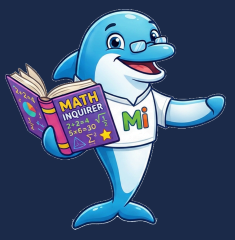
TUTOR GUIDE • LEVEL 1 • TERM 3 • LESSON 7 OF 10

Fill the Grid



Big idea: Each row and each column has one of each, so every empty box has a reason. In a grid puzzle each picture is used once in every row and once in every column. Look along the row and down the column to find what is missing.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Logic puzzle (not a Primary 1 syllabus topic) (Missing one in a row, and ordering by a rule (Primary 1 patterns, p. 31) are the nearest syllabus ideas; A grid in which every row and every column holds each picture once is a logic puzzle (a Latin square), not a syllabus topic; Elimination, the move used here, is in the Primary heuristics list (p. 13 to 14) as "make a systematic list and cross out") Strand label for the tutor only. NEW TOPIC, no Primary 1 school-paper evidence (the 23 Oct paper study found no grid item), so every word of the wording is marked unverified and this is the lesson that is safest to skip or swap (decision D13). BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only. Start here is approachable: one empty box in a row or a column with three pictures, a grid with one empty box in each row, and recognising a grid that follows the rule. Think harder (elimination in a 3 by 3 grid with only two clues, spotting a repeat, counting which shapes could go in a box, and what keeps a correct grid correct) is a logic move that the syllabus does not name. Go further is the beyond-syllabus item Four by Four: a 4 by 4 grid of the digits 1 to 4 solved by the same row-and-column reasoning, and what the rule says about a larger grid. The contest tag for the tutor is constraint grids (3 of 100 SASMO items, Section B). Owner decision, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item and a non-routine type (here LOG: elimination in a 3 by 3 grid with clues). Do not describe the lesson as MOE-aligned.
Objectives	• Say the rule of a grid puzzle: one of each picture in every row and one of each in every column. • Find the one missing picture in a row or in a column, and say whether the row or the column told us. • Use both the row and the column to find a box when neither one alone is enough. • Think harder: finish a grid from two clues, spot a repeat, say how many shapes could go in a box, and test what keeps a correct grid correct. • Go further: solve a 4 by 4 grid of the digits 1 to 4, and say how many of each digit a larger grid holds.
Joining cold	**Remember warm-up covers:** A row holds a circle, a square and a triangle once each: which one is missing from a row of two (spiral: Term 2 Lesson 1, find the missing one in a row, taught again with the tiles); Digits 1 to 4: which is missing from 1, 3 and 4 (counting to 4, taught again with a number row); One of each: which of two rows has a circle, a square and a triangle once each (the rule of the lesson, said aloud). Only counting to 4 and the names circle, square and triangle. Nothing from the lessons before is needed: the warm-up shows a row with one picture missing, a row of digits with one missing, and a pair of rows to compare. The words row, column and grid are taught in Explore and the Mini-lesson. The safest lesson of Term 3 to miss or swap: no later lesson needs it, and Mi's Challenge Quest has a gate on it that uses only Start here ideas (a 3 by 3 grid with five boxes filled). A child who misses it watches the recording, does workbook pages 1 and 2 and cuts out the tiles if they want to play. A child who joins here cold can follow: the warm-up re-teaches a missing one in a row, Explore is hands-on, and Start here keeps to three pictures and one empty box per row.
Child materials	• Your Term 3 book (Book 6), workbook pages 1 to 5, and the grid-tile page (page 6) with the 16 tiles already cut out by a grown-up. The two mats on that page are not cut • No printer or no cutting? Nine small things of three kinds (for example 3 buttons, 3 coins and 3 pasta shapes) on a 3 by 3 grid you draw on paper • Coloured pencils are optional • A pencil, a rubber and spare paper

Tutor materials	• The slide deck • Your own set of tiles to show on camera, if you wish • A visible clock No ruler is needed. Nothing needs to be bought. The tiles are small: they stay on the table, nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson: a grown-up cuts the tiles beforehand. In class we use 9 tiles (3 circles, 3 squares and 3 triangles) on mat 1; the digits on the tiles are for Go further.
Tiers	Start here (5 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Four by Four at G1, G2, G3: A 4 by 4 grid of the digits 1 to 4 solved by row-and-column elimination (G1 with nine clues, G2 with six), and what the rule says about a larger grid: a 5 by 5 grid holds five of each digit (G3). Not a syllabus topic; a logic puzzle. Named puzzle: Four by Four. T1 (two clues in a 3 by 3 grid) is the non-routine elimination item. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: logic: elimination in a 3 by 3 grid with clues; if it is not this and not that, it must be the other one; check every row and every column before saying the grid is finished; how many different grids fit the clues. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, with your own tiles. Check: S1 circle, triangle, square; S2 true, false, true; S3 triangle and the column; S4 No; S5 3; T1 triangle, circle, triangle; T2 2 and 1; T3 true, false; G1 4, 1, 2, 1, 4, 3, 1; G2 3, 4, 2; G3 25 and 10.
- Tell parents a week before: the strip page for Lesson 8 of Term 3 is the next one to cut. For this lesson, tear out the grid-tile page (page 6) along the dashed line; a grown-up cuts out the 16 tiles and keeps them in an envelope; the two mats are not cut. With no printer: nine small things of three kinds on a 3 by 3 grid drawn on paper.
- Have the slides open. Have your own tiles to show on camera if you wish. Say the words row and column the same way every time, and keep the sentence frame: the row has ... and ..., so the box must be .... Do not teach a corner-first rule.
- Check the recording captures the tutor and the shared screen. Children video is excluded; any unmuted voice is recorded. Greet the whole group warmly without first names, then say the recording notice; do not use first names while the recording runs. Nobody holds paper up to the camera.
- Be plain with families: finding the missing one in a row is early Primary 1 work; solving a whole grid by elimination, the 4 by 4 grid of digits and what a bigger grid holds go beyond Primary 1 on purpose. This is a new topic with no Primary 1 school-paper evidence: the wording is unverified, and this is the lesson the tutor may swap or skip. Do not call the lesson MOE-aligned.
- Slides s-m3 and s-m5 keep the answers off the screen while you ask: the answer slide follows. Read every question, every option of E2 and the box labels aloud: the shapes on the slides are small.

Safeguarding and recording (this lesson)

- The tiles are small: they stay on the table, nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson; a grown-up cuts the tiles beforehand.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Making shapes with the arms is quiet and small; a child may stay seated and make the shapes with their hands.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Fill the Grid	1:33
2–8	**Remember warm-up**	6	3: Warm-up: which is missing?, Which digit is missing?, Which row has one of each?	4:53
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:23
12–23	**Explore**	11	3: Explore: one of each in every row, Explore: one of each in every column, Explore: the same top row	8:36
23–34	**Mini-lesson**	11	6: The rule, Look along the row. Look down the column., Use the row and the column, Is the grid finished?, Your turn, Check: your turn	8:31
34–38	**Brain break**	4	1: Be the shapes	3:22
38–70	**Workbook**	32	6: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Workbook page 2: Start here, Check page 2, Choose your next page	26:03
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:26
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:42
82–85	**Look-back**	3	1: Look-back	2:23
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:44
	Total	**90**		**68:36**

The script needs 68:36 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 81:08. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 38 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 38, skip in this order: Start with: shorten Explore round 3 (how many grids) to a quick look and carry it to the Mini-lesson; Then: cut the second item of the Your turn slide (s-m5); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:33 of 2:00

0:00	SLIDE	Show slide **Fill the Grid** (s-title)
0:15	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your grid tiles. Pause or rewind any time.*
0:26	SAY	Hello, inquirers! Pencil ready, and your grid tiles and mat on the table. Cameras can be on or off, as you like.
0:37	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:54	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:10	SAY	Today we play with a grid. A grid is a square of boxes. We fill it so that every box has a reason.
1:21	DO	Greet the whole group warmly. Do not use children's first names on the recording. [0:12]
	NOTE	Keep this to two minutes. Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. The routine and the four inquirer rules were introduced in the first lesson of Term 1: do not re-teach them.
	NOTE	Plain statement for families when it comes up: finding the missing one in a row is early Primary 1 work; solving a whole grid by elimination, the 4 by 4 grid of digits and what a bigger grid holds go beyond Primary 1 on purpose, always with the tiles in hand. This is a new topic with no school-paper evidence, so the wording is unverified.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. A child without the tiles draws a 3 by 3 grid on paper and writes C for circle, S for square and T for triangle.

2:00 – 8:00 Remember warm-up

script 4:53 of 6:00

2:00	SLIDE	Show slide **Warm-up: which is missing?** (s-rem1)
2:25	REC	*Recording: join in! Find the missing shape and the missing digit, and say which row has one of each. Pause if you need longer.*
2:37	SAY	A row has one of each: a circle, a square and a triangle. Here is a row with a circle and a square. One shape is missing.
2:50	ASK	Which shape is missing from the row? [wait about 15 s for answers]
3:09	ASK	How do you know? [wait about 20 s for answers]
3:31	CHILDREN	Children put the circle tile and the square tile in a row, then add the missing tile. [0:30]
4:01	SLIDE	Show slide **Which digit is missing?** (s-rem2)
4:26	SAY	Now numbers. The digits one, two, three and four are used once each in a row. Here are one, three and four, and a mark.
4:38	ASK	Which digit is missing? [wait about 15 s for answers]
4:56	SLIDE	Show slide **Which row has one of each?** (s-rem3)
5:20	SAY	One of each means a circle, a square and a triangle, once each. Look at row A and row B.

5:30	**ASK**	Which row has one of each? [wait about 15 s for answers]
5:48	**ASK**	What is wrong with the other row? [wait about 20 s for answers]
6:12	**CHILDREN**	Children put three tiles in a row with one of each, then a row with a repeat, and say which is which. [0:35]
6:47	**SAY**	Remember those three words: one of each. We use them all lesson.
	IF STUCK	If a child is unsure of a shape name, point to the tile or the picture and say its name. For the digits, count one, two, three, four aloud and notice the skipped number. For one of each, count the three kinds on fingers.
	NOTE	Answers: triangle (the row has a circle and a square); 2; row A (row B has two squares and no circle). Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 4 and the three shape names.

8:00 – 12:00 Sneak-peek puzzles

script 3:23 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	**READ**	Puzzle one. Each row has a circle, a square and a triangle, one of each. This row has a square and a triangle. Which shape goes in the empty box? Write its name.
9:05	**CHILDREN**	Children write the shape name on their own paper. Tiles are fine to use. [0:40]
9:45	**READ**	Puzzle two. Every row and every column has a circle, a square and a triangle, one of each. Which shape goes in the first box with a question mark in the bottom row? I know because blank.
10:05	**CHILDREN**	Children write or draw their answer. [1:10]
11:15	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: circle. Puzzle 2 answer: a square (the column of box a has a triangle and a circle; the row shows only a triangle). Look for good ideas to praise later (a child who looked down the column, or who said it cannot be a triangle or a circle). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 8:36 of 11:00

12:00	**SLIDE**	Show slide **Explore: one of each in every row** (s-explore1)
12:25	**REC**	*Recording: pause now. Put one of each in every row, then in every column, with your tiles. Press play when you have tried.*
12:36	**SAY**	Today your tiles are the teacher. Put mat one in front of you. Take three circles, three squares and three triangles. Put the stars and the digits away.
12:50	**SAY**	A row goes across, like children in a queue. Fill the mat so that every row has a circle, a square and a triangle. Only the rows, for now.
13:05	**CHILDREN**	Children fill mat 1 so every row has a circle, a square and a triangle. Tutor scans the cameras and the tables. [1:10]
14:15	**ASK**	What is in your top row? Tell me the three shapes. [wait about 25 s for answers]
14:46	**SLIDE**	Show slide **Explore: one of each in every column** (s-explore2)
15:10	**SAY**	Round two. A column goes down, from the top to the bottom. Now look down each column. Is there one of each? If not, move your tiles until every row and every column has one of each.
15:29	**CHILDREN**	Children fix their grid. Many different grids are right. [1:50]
17:19	**ASK**	How did you fix a column that was wrong? [wait about 25 s for answers]
17:48	**SLIDE**	Show slide **Explore: the same top row** (s-explore3)

18:14	SAY	Round three. Keep this top row: circle, square, triangle. Fill the other two rows. Every row and every column has one of each. How many different grids can you make?
18:28	CHILDREN	Children make one grid, then look for a different one with the same top row. [1:20]
19:48	ASK	How many different grids did you find? [wait about 15 s for answers]
20:07	ASK	How do you know there is not another one? [wait about 25 s for answers]
	IF STUCK	A child who has two of a shape in a column: ask them to read down the column aloud. A child who is stuck on round two: keep the top row as it is and ask where the circle can go in the middle row. A child with too few tiles uses the letters C, S and T on paper.
	NOTE	Main room only; children say what they have made. Tilting the camera to the table is allowed; a face need not be in view; nothing is held up to the camera. The answer to round three is 2 grids: circle square triangle then square triangle circle then triangle circle square; or circle square triangle then triangle circle square then square triangle circle. Do not say the rule before the children do. Do not confirm the number 2 until the next slide.

23:00 – 34:00 Mini-lesson

script 8:31 of 11:00

23:00	SLIDE	Show slide **The rule** (s-m1)
23:25	REC	*Recording: keep watching and copy the examples with your tiles. Rewind if you miss a step.*
23:33	SAY	Here is the rule. One of each in every row. One of each in every column. This grid follows the rule. Read the top row with me: square, circle, triangle.
23:48	ASK	Is there one of each in the first column, going down? [wait about 15 s for answers]
24:08	SAY	With the top row circle, square, triangle there were exactly two grids. This one has a different top row, so it is a new grid.
24:21	SLIDE	Show slide **Look along the row. Look down the column.** (s-m2)
24:46	SAY	Look along the row. Then look down the column. A column is just a row that stands up.
24:55	ASK	The row has a triangle, then a mark, then a square. Which shape is missing? [wait about 15 s for answers]
25:18	ASK	The column has a mark, then a circle, then a triangle. Which shape is missing? [wait about 15 s for answers]
25:40	SAY	Say it with me: the row has a triangle and a square, so the box must be a circle.
25:50	SLIDE	Show slide **Use the row and the column** (s-m3)
26:14	SAY	Now a box that needs both. Look at box a, in the middle. Its row has a square. Its column has a circle.
26:26	ASK	Could box a be a square? [wait about 12 s for answers]
26:41	ASK	Could box a be a circle? [wait about 12 s for answers]
26:56	ASK	So what shape is left for box a? [wait about 15 s for answers]
27:15	SAY	If it is not this and not that, it must be the other one. A triangle, because the row stops a square and the column stops a circle.
27:29	CHILDREN	Children may copy the grid onto mat 1 with their tiles and check. [0:30]
27:59	SLIDE	Show slide **Is the grid finished?** (s-m4)
28:24	SAY	A grid is not finished until we check every row and every column. Look at the rows first.
28:33	ASK	Does every row have one of each? [wait about 15 s for answers]
28:52	SAY	Now the columns. Run a finger down each one.
28:56	ASK	Which columns have two of the same? [wait about 20 s for answers]
29:20	SAY	So this grid is not correct. We never check only the rows.
29:26	SLIDE	Show slide **Your turn** (s-m5)

29:50	SAY	Your turn. On your paper: which shape goes in the row? And is the grid correct, yes or no? Check the rows and the columns.
30:03	CHILDREN	Children write the two answers on their own paper. Wait 10 seconds before helping. [0:55]
30:58	SLIDE	Show slide **Check: your turn** (s-m5a)
31:23	READ	Row: a triangle. Grid: yes, it is correct. Tick your own work with a tick.
	NOTE	Key words to use every time: row, column, one of each, grid, check. Say it the same way each time: the row has ... and ..., so the box must be .... Do not use the word wrong for a child, only for a grid or a row.
	IF STUCK	If a child checks only the rows, ask them to touch the first column from the top. If a child is unsure which shape is missing, put the three tiles in a line and remove the two that are there.

34:00 – 38:00 Brain break

script 3:22 of 4:00

34:00	SLIDE	Show slide **Be the shapes** (s-break)
34:25	REC	*Recording: stand up and join in! Make the shapes with your body as I call them.*
34:33	SAY	Stand up and get a sip of water. I call a shape and you make it with your arms or your body. A circle: a big ring. A square: elbows out. A triangle: a roof above your head.
34:52	CHILDREN	Call circle, square and triangle one at a time, then a row of three (circle, square, triangle), then a different row. Children make each shape; a child who sits makes the shapes with their hands. [2:30]

38:00 – 70:00 Workbook

script 26:03 of 32:00

38:00	SLIDE	Show slide **Workbook page 1: Start here** (s-ws1)
38:25	REC	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
38:34	READ	Page 1, Start here. Question 1: each row has a circle, a square and a triangle. Which shape goes in the box with the mark in rows A, B and C? Circle one for each. Question 2: here are three rows. Tick true or false for each statement: this row has one of each.
39:03	CHILDREN	Page 1, questions 1 and 2. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [7:30]
46:33	SLIDE	Show slide **Check page 1** (s-ws1ans)
46:58	READ	Question 1: row A, circle; row B, triangle; row C, square. Question 2: row A true, row B false, row C true. Tick your own page with a tick.
47:14	SAY	After question 2 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
47:23	SLIDE	Show slide **Workbook page 2: Start here** (s-ws2)
47:48	READ	Page 2. Question 3: look at the grid. Circle the shape for the bottom left box. Then circle the reason that tells us. Question 4: look at the grid in the picture. Every row has one of each. Is the grid correct? Circle Yes or No.
48:13	SLIDE	Show slide **Workbook page 2: Start here** (s-ws2b)
48:38	READ	Question 5: every row of a 3 by 3 grid has one circle, and every column has one circle. How many circles are in the whole grid?
48:53	CHILDREN	Page 2, questions 3 to 5. [9:00]
57:53	SLIDE	Show slide **Check page 2** (s-ws2ans)
58:18	READ	Question 3: a triangle, because the column has a circle and a square. Question 4: look at the grid in the picture. Every row has one of each. Is the grid correct? Circle Yes or No. Question 5: every row of a 3 by 3 grid has one circle, and every column has one circle. How many circles are in the whole grid?
58:52	SLIDE	Show slide **Choose your next page** (s-choice)
59:17	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.

59:33	**MOVE**	Stand and reach up for 30 seconds, then sit down and choose.
60:03	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [4:00]
	IF STUCK	Q1: which two shapes are in the row? Q2: touch each shape in the row and check there is a circle, a square and a triangle. Q3: look down the column of the bottom left box, then at its row. Q4: build the grid with tiles, then read down each column. Q5: how many rows? one circle in each.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder has two clues, how many shapes could go in a box and what keeps a correct grid correct. Go further has Four by Four, a digit grid, and a bigger grid (beyond Primary 1 on purpose). Do not announce who is on which page. Start here takes a middle child about 15 minutes, so the choice time is short: for most children pages 3 to 5 are for home. Say so plainly, do not skip the answer slides, and take any extra minutes from the flex time.

70:00 – 76:00 Talk time

script 4:26 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:25	**REC**	*Recording: pause. Tell a grown-up whether the child is right, and test it with your tiles.*
70:33	**READ**	A child says: I can fill every empty box by looking along the row. I never need the column. Is the child right? Test it with your tiles.
70:48	**CHILDREN**	Test it with your tiles: set up the grid from the sneak-peek and look at the bottom row, which has two empty boxes. [0:45]
71:33	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:13	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 70 s for answers]
73:29	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
73:54	**SAY**	Remember the sneak-peek? Puzzle one: a circle. Puzzle two: a square, because the column has a triangle and a circle. The row could not tell us alone.
74:08	**ASK**	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: not always. If a row has only one empty box, the row tells us. If a row has two empty boxes, the row cannot decide alone, so we look down the column too. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:42 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:25	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	**READ**	E1. A row has a circle and a square. Which shape is missing from the row? Circle one. E2. Which row has one of each? Choose the correct answer. Option 1: circle, circle, square. Option 2: square, triangle, circle. Option 3: triangle, triangle, circle. Option 4: circle, square, square. Write its number in the bracket. E3. A box has a circle in its row and a square in its column. Which shape goes in the box? Circle one.
77:24	**CHILDREN**	Children answer E1 to E3 on their own. [2:45]
80:09	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:34	**READ**	E1: triangle. E2: option 2. E3: triangle.
80:37	**SAY**	Tick your own ticket with a tick. Well done.

	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:23 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
83:03	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:44 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Make a Table. Please have the strip page cut out by a grown-up, fourteen paper strips, and a pencil. No matches or toothpicks, please. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: three in a row** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: a grid of one, two and three with one, two and three on the long line from the top left to the bottom right. Fill it so every row and every column has one, two and three once.
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: top row 1, 3, 2; then 3, 2, 1; then 2, 1, 3. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	A child says: if every row has one of each, then every column has one of each too. Is that always true? Test it with your tiles: put the same row three times.	*Recording: pause. Is one of each in every row always one of each in every column? Test it with three tiles in the same order.*	No. Three rows that are all circle, square, triangle have one of each in every row, but every column has three of the same shape. We must check the columns too.

Make a puzzle for your grown-up	4 (optional)	Fill mat 1 with a correct grid. Cover three boxes with small pieces of paper. Ask a grown-up to say what is hidden and how they know. Check with the tiles.	*Recording: pause. Fill a grid. Cover three boxes with paper. Ask a grown-up to find the hidden shapes and say why.*	No single answer. A puzzle works when every covered box has a reason from its row or its column. If too many boxes are covered, two different grids may fit, and the puzzle has no one answer.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never seen a grid puzzle. Teach them how to find a missing shape: look along the row, then look down the column.	*Recording: pause. Teach a grown-up how to find a missing shape by looking along the row and down the column.*	Look along the row to see which shapes are there. Look down the column to see which shapes are there. The shape that is in neither is the one that goes in the box.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): A row needs a circle, a square and a triangle, one of each. This row has a square and a triangle. Which shape goes in the empty box? Write its name.

> **Answer: circle**

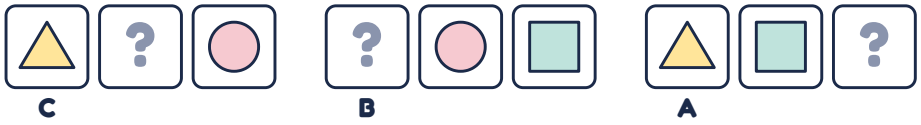
Puzzle 2 (think): Every row and every column has a circle, a square and a triangle, one of each. Which shape goes in the first ? box of the bottom row? I know because ___.

> **Answer: A square. The first box in the bottom row has a triangle and a circle above it in its column, so it must be a square. The row has only a triangle, so the column decides.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Each row has a circle, a square and a triangle. Which shape goes in the box with the ? mark in rows A, B and C?**



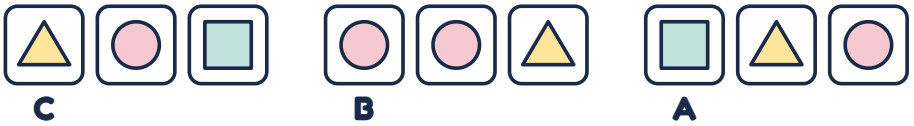
(a) Row A. Circle one.
(b) Row B. Circle one.
(c) Row C. Circle one.

> **Answer: circle; triangle; square**

Hint (say this first): Which two shapes are in the row already? Which one is not there?
Worked solution: Row A has a triangle and a square, so the circle is missing. Row B has a circle and a square, so the triangle is missing. Row C has a triangle and a circle, so the square is missing.
Common mistakes: Choosing a shape that is already in the row.; Circling without looking at the other two shapes.
Time for a middle child: about 3 min.

True/False statements: Row A has one of each.; Row B has one of each.; Row C has one of each.

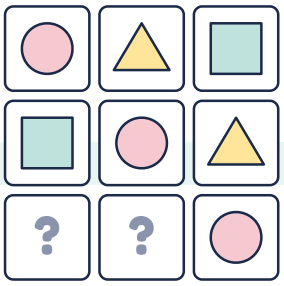
S2 **Look at the three rows. Tick True or False for each statement.**



> **Answer: A true; B false; C true**

Hint (say this first): One of each means a circle, a square and a triangle, once each. Touch the shapes one at a time.
Worked solution: Row A has a square, a triangle and a circle: one of each, so True. Row B has two circles and no square, so False. Row C has a triangle, a circle and a square: one of each, so True.
Common mistakes: Saying True when a shape is used twice.; Looking at only the first two shapes.
Time for a middle child: about 3 min.

S3 **Every row and every column has a circle, a square and a triangle. Look at the bottom left box.**

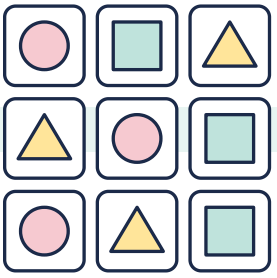


(a) Which shape goes in the bottom left box? Circle one.
(b) Which reason tells us? Circle one.

Answer: triangle; the column has a circle and a square

Hint (say this first): Look down the column of the bottom left box. Then look along its row. Which one has two different shapes?
Worked solution: The column of the bottom left box has a circle at the top and a square in the middle, so the triangle is missing. The row has only a circle, so the row could not tell us.
Common mistakes: Choosing the row reason: the row has only one shape.; Choosing a shape that is already in the column.
Time for a middle child: about 3.5 min.

S4 **A child fills the grid in the picture. Every row has one of each. Is the grid correct?**



Circle Yes or No.

Answer: No

Hint (say this first): Check the columns too. Run a finger down the first column. Use your tiles to build the grid.
Worked solution: The first column is circle, triangle, circle: two circles. So the grid is not correct, even though every row has one of each.
Common mistakes: Saying Yes after checking only the rows.; Not building the grid to look at the columns.
Time for a middle child: about 3 min.

S5 **Every row of a 3 by 3 grid has one circle, and every column has one circle. How many circles are in the whole grid?**

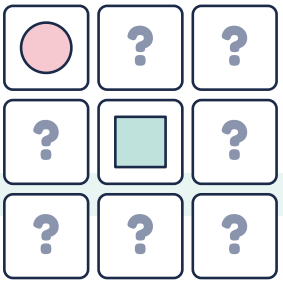
There are ____ circles in the grid.

Answer: 3

Hint (say this first): How many rows are there? Each row has one circle.
Worked solution: There are 3 rows and each row has one circle, so there are 3 circles. (There are 3 columns with one circle each, which is the same 3 circles.)
Common mistakes: Writing 9, the number of boxes.; Writing 1, the circles in one row.
Time for a middle child: about 2 min.

Think harder (page 3): the taught idea in a new setting

T1 **Mi has a grid with only two shapes in it. Every row and every column has one of each. Which shape goes in each box?**



(a) Top middle box.
(b) Middle right box.
(c) Bottom right box.

Answer: triangle; circle; triangle

Hint (say this first): Start with the top middle box. Its row has a circle. Its column has a square. What is left?
Worked solution: The top middle box is in the top row with a circle and above the square, so it cannot be a circle or a square: it is a triangle. Then the top row is circle, triangle, so the top right is a square. The middle row has a square, and its right box sits under a square, so the middle row is triangle, square, circle: the middle right box is a circle. The bottom row is square, circle, triangle, so the bottom right box is a triangle.
Common mistakes: Guessing a shape that is already in the row or the column.; Looking only along the row: the top middle box has two empty boxes in its row.
Time for a middle child: about 5 min.

T2 **Every row and every column of a grid has a circle, a square and a triangle. How many different shapes could go in each box?**

(a) Circle in its row, circle in its column: ☐ shapes.
(b) Triangle in its row, square in its column: ☐ shape.

Answer: 2; 1

Hint (say this first): Cross out the shapes that are already in the row or in the column. Use your tiles to see it.
Worked solution: First box: the circle is crossed out, so a square or a triangle could go in: 2. Second box: the triangle is crossed out by the row and the square by the column, so only the circle is left: 1.
Common mistakes: Counting the shapes that are already there instead of the shapes that could go in.; Forgetting that the row and the column can show the same shape.
Time for a middle child: about 3.5 min.

True/False statements: Swap the top row and the bottom row. The new grid is still correct.; Swap two boxes in the middle row. The new grid is still correct.

T3 **A grid is correct: every row and every column has one of each. A child changes it. Test each change with your tiles. Tick True or False.**

Answer: first true; second false

Hint (say this first): Try it with your tiles. After a swap, check the rows and then the columns.
Worked solution: Swapping two whole rows keeps one of each in every row, and every column still holds the same three shapes, so True. Swapping two boxes inside one row keeps that row correct, but the columns change and one column then has a repeat, so False.
Common mistakes: Checking only the rows after a swap.; Thinking that any swap breaks the grid.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Four by Four. Every row and every column has the digits 1, 2, 3 and 4 once each. Write the digits in the empty boxes.**

1		2	3
3		4	2
	3		4
	2		

Answer: 4, 1, 2, 1, 4, 3, 1 (reading the empty boxes row by row)

Hint (say this first): Find a row or a column with only one empty box. Fill it. Then look for the next one.
Worked solution: The grid is 1 4 2 3, 3 1 4 2, 2 3 1 4 and 4 2 3 1. For example, the third box of the bottom row: its column has 2, 4 and 1, so it is 3. The first box in the bottom row: its row has 2 and 3 and 1, so it is 4.
Common mistakes: Repeating a digit in a column.; Filling a box by looking only at the row when it has two empty boxes.
Time for a middle child: about 6 min.

G2 **Four by Four, with fewer clues. Only six digits are given. Every row and every column has 1, 2, 3 and 4 once each. Which digit goes in each box marked a, b and c? The other empty boxes are not asked.**

2	1		a
	4		
b	3	1	
	c	3	

Box a is ____ . Box b is ____ . Box c is ____ .

Answer: 3; 4; 2

Hint (say this first): Find a box with only one digit that can go in. Fill it, then use the new digit.
Worked solution: The grid is 2 1 4 3, 3 4 2 1, 4 3 1 2 and 1 2 3 4. For example, the first box of the third row: its row has 3 and 1, so it is 2 or 4; its column has 2 and 3 and 1 after we find them, so it is 4.
Common mistakes: Guessing a digit that is already in the row.; Stopping at the first digit that fits without checking the column.
Time for a middle child: about 6 min.

G3 **A 5 by 5 grid uses the digits 1, 2, 3, 4 and 5. Every row and every column has each digit once.**
(a) How many boxes are in the grid? ____
(b) How many boxes hold a 4 or a 5? ____

> **Answer: 25; 10**

Hint (say this first): A row has 5 boxes. How many rows are there? For part b: each row has one 4 and one 5.
Worked solution: There are 5 rows of 5 boxes: 5, 10, 15, 20, 25, so 25 boxes. Each row has one 4 and one 5, so 2 boxes in each row: 5 rows of 2 make 10 boxes that hold a 4 or a 5.
Common mistakes: Writing 5 for the boxes in all (one row).; Writing 5 or 2 for the boxes that hold a 4 or a 5.
Time for a middle child: about 4 min.

Investigation (page 5): Grid builder

Easy first step: Row 1 is 1 3 2. Row 2 is 3 2 1. Write row 3: ____ ____ ____
Then: Put a 2 in the middle of a 3 by 3 grid. Every row and column has 1, 2 and 3 once. Write each grid you can make on its own line, row by row. The first one is 132 321 213. A turned grid is a different grid. How many are there?

> **Model answer: First: row 3 is 2, 1, 3. With a 2 in the middle there are exactly 4 grids: 1 3 2 / 3 2 1 / 2 1 3; 2 1 3 / 3 2 1 / 1 3 2; 2 3 1 / 1 2 3 / 3 1 2; and 3 1 2 / 1 2 3 / 2 3 1. The task says a turned grid is a different grid (any box different), so a grid turned upside down or swapped left to right counts again, and the answer is 4. A child who finds 2 can often find the other 2 by turning one upside down or swapping it left to right.**

Look for: A child who tries a corner, checks each row and each column, and then swaps or turns a grid to get a new one; a child who notices that the three 2s always run along one of the two diagonals (two choices) and that each choice gives two grids; a child who says why there cannot be a fifth. Credit any clear reason. Do not mark a child down for stopping at one or two grids.
Common mistakes: Repeating a digit in a column.; Writing the same grid twice.

Talk and explain (minutes 70 to 76)

A child says: I can fill every empty box by looking along the row. I never need the column. Is the child right? Test it with your tiles. Sentence starter: I think ... because ...

> **Model answer: Not always. If a row has only one empty box, the row tells us the missing shape. If a row has two empty boxes, the row cannot decide alone, so we look down the column too. We can test it with the bottom row of the sneak-peek grid.**

Look for: A child who tests it with the tiles, or says the row has two empty boxes so the row cannot decide, or says we need the column when a row has more than one empty box. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **A row has a circle and a square. Which shape is missing from the row? Circle one.**

> **Answer: triangle**

Worked solution: The row has a circle and a square. The triangle is missing, so the triangle goes in the box.
Common mistakes: Choosing a shape that is already in the row.; Circling without looking at the row.

E2 **Which row has one of each? Choose the correct answer. Write its number in the bracket.**

> **Answer: (2) square, triangle, circle**

Worked solution: Row (2) has a square, a triangle and a circle: one of each. Row (1) has two circles, row (3) has two triangles and row (4) has two squares.
Common mistakes: Choosing (1) because it starts with a circle.; Not checking every shape in the row.

E3 **A box has a circle in its row and a square in its column. Which shape goes in the box? Circle one.**

Answer: triangle

Worked solution: Box a cannot be a circle, because the row has a circle. It cannot be a square, because the column has a square. The triangle is left.
Common mistakes: Choosing the circle or the square.; Looking at the row only.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (finding the missing shape; using the row and the column; checking the whole grid; something else); *What helped me* (my tiles; looking along and down; checking my answer; I was not stuck); *Next time I will try* (using my tiles; checking every column; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	A row has a circle and a triangle. The row must have one of each: a circle, a square and a triangle. Which shape is missing? ____	square
H2	Start here	In a column the digits 1, 2 and 3 are used once each. Going down, the column shows 3, ?, 1. What is the ? ____	2
H3	Think harder	A 3 by 3 grid uses 1, 2 and 3. Draw a 3 by 3 grid of boxes. Row 1 is 1, 2, 3. Row 2 starts with 2. Write rows 2 and 3. Row 2: ___ ___ ___ Row 3: ___ ___ ___	row 2: 2 3 1; row 3: 3 1 2
H4	Go further	A 4 by 4 grid uses 1, 2, 3 and 4. Draw a 4 by 4 grid of boxes. Row 1 is 1, 2, 3, 4. Row 2 is 2, 1, 4, 3. Row 3 starts with 3. What is the second box of row 3? ____	4
H5	Start here	Count up from 46 to 52. How many steps is that? ____	6

Online practice (`out/l27/practice`, key `t3l27`): 11 items; 4 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to use clues to work out what must go in an empty box. This week: filling grids so every row and every column has one of each.

Term line (plan, section 5): Your child is learning to give a reason for each step. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Fill the Grid** s-title	TODAY WE WILL: / Fill a grid so every row and column has one of each.	Lesson 7 of Term 3, 90 minutes. Objective: say the rule of a grid puzzle (one of each in every row and every column), find a missing picture from its row or its column, and use both together. Plain statement for families: finding the missing one in a row is early Primary 1 work; solving a whole grid by elimination, the 4 by 4 grid of digits and what a larger grid holds go past the school Primary 1 syllabus on purpose. This is a new topic with no school-paper evidence, so it is the safest lesson of the term to miss. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 34, brain break 34 to 38, workbook 38 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	2–8	**Warm-up: which is missing?** s-rem1	A row has one of each: a circle, a square and a triangle. / [picture: panel] / A circle and a square are here. Which shape is missing?	A row with a circle and a square and a ? mark. Children say which shape is missing (triangle) and touch the tile of that shape in their kit if they have one. This is the missing-one idea taught again with pictures: three kinds, one of each. Ask: how do you know? (a circle and a square are here, so the triangle is missing). The slide never shows the answer. A child who is unsure counts the three kinds on their fingers. No writing, no marks.
3	2–8	**Which digit is missing?** s-rem2	The digits 1, 2, 3 and 4 are used once each. / 1 3 4 ? / Which digit is missing?	A row of digits 1, 3, 4 and a ? mark. The digits 1 to 4 are used once each, so the missing digit is 2. Children whisper or show fingers. This is the same idea with numbers, and it is the first look at the 4 by 4 grid that comes at Go further. A child who is unsure counts 1, 2, 3, 4 aloud and notices which number was skipped. Do not ask for a written answer.
4	2–8	**Which row has one of each?** s-rem3	One of each means a circle, a square and a triangle, once each. / [picture: panel] / [picture: panel] / Which row has one of each?	Row A is circle, square, triangle: one of each. Row B is square, triangle, square: two squares and no circle. Ask: which row has one of each (row A); what is wrong with row B (two squares, no circle). Say the rule in your own words: one of each, no repeats, nothing missing. Children answer with a thumb or by naming the row. Nobody writes. Joining cold: all a new child needs is the names of the three shapes.
5	8–12	**Sneak-peek puzzles** s-sneak	PUZZLE 1: anyone can try: / Each row has one of each. Which shape goes in the empty box? / [picture: panel] / PUZZLE 2: think: / Which shape goes in the first ? of the bottom row? I know because ___. / [picture: panel]	Puzzle 1 (a row with a square and a triangle: the circle is missing) is an access question: any child can see which of three is missing. Puzzle 2 is a thinking question on the lesson idea: the first ? in the bottom row is the box we ask about; its column holds a triangle and a circle, so it must be a square, while its row shows only a triangle, so the row alone cannot decide. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: circle; square. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
6	12–23	**Explore: one of each in every row** s-explore1	Put mat 1 in front of you. / Take 3 circles, 3 squares and 3 triangles. / Fill the mat. Every row has a circle, a square and a triangle. / Rows only, for now.	Round 1, about 3 minutes. Children use mat 1 and 9 tiles (3 circles, 3 squares, 3 triangles; the stars and the digits are not used today). Say the words: a row goes across, like a line of children in a queue. Every row must have a circle, a square and a triangle. Most children will make three rows that are the same: that is fine, it follows the rule so far. Take volunteers: what is in your top row. Do not say the column rule yet. A child with no tiles draws three rows on paper, writing C, S and T. Do not name a child with a wrong answer.

7	12–23	**Explore: one of each in every column** s-explore2	A column goes down. / Look down each column. Is there one of each? / Move tiles until every row and every column has one of each. / Take your time. Many ways are right.	Round 2, about 4 minutes, main room. Now the same rule goes down: a column is a line of boxes from the top to the bottom. Children check each column, find a repeat (two circles in one column), and swap tiles. Give one hint only when asked: leave the top row as it is and look at where the circle can go in the middle row. Take volunteers: how did you fix it. Several grids are right. Children say the rule in their own words before moving on: one of each in every row and every column. A child with no tiles uses the three letters on paper.
8	12–23	**Explore: the same top row** s-explore3	[picture: panel] / Keep this top row. / Fill the other two rows. Every row and every column has one of each. / How many different grids can you make? / Make one. Then make a different one.	Round 3, about 3 minutes. Everyone keeps the top row circle, square, triangle on mat 1. How many different grids can they fill? There are exactly 2: the middle row can start with a square (square, triangle, circle) or with a triangle (triangle, circle, square), and then the bottom row is fixed. Children find one, then look for another. Ask: how many different grids did you find; how do you know there is not a third. The answer 2 is not on the slide and is confirmed on the next slide. A child who finds a grid that breaks the rule is asked to check each column.
9	23–34	**The rule** s-m1	THE RULE: / One of each in every row. / One of each in every column. / [picture: panel]	Name the rule and show a finished grid. A row goes across, a column goes down. Read each row of the grid on the slide: square, circle, triangle; triangle, square, circle; circle, triangle, square. Then each column: square, triangle, circle; circle, square, triangle; triangle, circle, square. Every one has one of each. Also confirm the answer to Explore round 3: with the top row circle, square, triangle there were 2 grids. This grid has a different top row, so it is a new one. Say: a grid like this is a puzzle square, and every empty box in a puzzle has a reason.
10	23–34	**Look along the row. Look down the column.** s-m2	ALONG THE ROW: / What is missing? / [picture: panel] / DOWN THE COLUMN: / What is missing? / top box: ? / middle box: circle / bottom box: triangle	Two quick worked examples. Along the row: triangle, ?, square, so the circle is missing. Down the column: a mark at the top, then a circle, then a triangle, so the square is missing. Ask each question first; children answer aloud or touch the tile. Then say: a column is just a row turned to stand up. Say the sentence frame: the row has a triangle and a square, so the box must be a circle. Children say the frame with you. The slide never shows the answers.
11	23–34	**Use the row and the column** s-m3	[picture: panel] / BOX a IS THE MIDDLE BOX: / Its row has a square. / Its column has a circle.	The worked elimination. Box a is in the middle of the grid. Its row has a square, so box a is not a square. Its column has a circle, so box a is not a circle. It cannot be a circle or a square, so it must be a triangle. Say the move slowly: if it is not this and not that, it must be the other one. Ask: could it be a square (no, the row has a square); a circle (no, the column has a circle); so what is left (triangle). Children test with their tiles on mat 1 if they set up the same grid. The slide does not show the answer.
12	23–34	**Is the grid finished?** s-m4	[picture: panel] / Check every row. Check every column. Is it correct?	The check move: a grid is not finished until every row and every column has one of each. Rows: circle, square, triangle (yes); square, triangle, circle (yes); circle, triangle, square (yes). Columns: circle, square, circle (no: two circles); square, triangle, triangle (no); triangle, circle, square (yes). So the grid is not correct. Ask: do all the rows follow the rule (yes), so are we done (no, we must check the columns too). Children say which columns have a repeat (the first and the second). Never check only the rows.
13	23–34	**Your turn** s-m5	ROW: / Which shape goes in the ? box? / [picture: panel] / GRID: / Is the grid correct? Circle Yes or No. / [picture: panel]	Two quick items, answered on the children's own paper (say so aloud): a row with a square and a circle (the triangle is missing), and a finished grid to check (it is correct: every row and every column has one of each). Walk the cameras and wait 10 seconds before helping. Nobody holds paper up. The next slide shows the answers and children tick their own work. Answers: triangle; Yes.

14	23–34	**Check: your turn** s-m5a	Row: a triangle. Grid: Yes, it is correct.	Read both answers aloud so a child watching the recording can tick their own work. The row has a square and a circle, so the triangle is missing. The grid: rows are square, triangle, circle; triangle, circle, square; circle, square, triangle, and the columns are square, triangle, circle; triangle, circle, square; circle, square, triangle, so every row and every column has one of each. Never ask a child to show a wrong answer to the group. A child who says No is asked to check the columns with a finger.
15	34–38	**Be the shapes** s-break	Stand up. I call a shape. / Make it with your arms or your body. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Call the shapes one at a time: a circle (arms in a big ring above the head), a square (arms out to the sides and elbows bent at right angles), a triangle (hands together above the head like a roof). Then call a row of three, for example circle, square, triangle, and children make the three shapes one after another. Call one more row with a different order. No equipment, no running; a child may stay seated and make the shapes with their hands. Do not give prizes or compare children.
16	38–70	**Workbook page 1: Start here** s-ws1	Questions 1 and 2. Take your time. / questions S1, S2	Read questions 1 and 2 aloud. Children work on their own with their tiles; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 2 any child may choose to carry on or try Think harder. Question 1: the missing shape in three rows. Question 2: tick True or False for three rows.
17	38–70	**Check page 1** s-ws1ans	answers S1, S2	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 circle, triangle, square; Q2 row A true, row B false, row C true.
18	38–70	**Workbook page 2: Start here** s-ws2	Questions 3 and 4. Take your time. / questions S3, S4	Read questions 3 and 4 aloud, then show the next slide for question 5. Question 3: the bottom left box, and the reason that tells us (the column has the clue). Question 4: a grid in a picture, every row correct, is it correct (the columns). Children may use their tiles to check. Wait 10 seconds before helping.
19	38–70	**Workbook page 2: Start here** s-ws2b	Question 5. Take your time. / questions S5	Question 5: how many circles in a whole 3 by 3 grid. Children may use their tiles to check. Wait 10 seconds before helping. Keep this slide up while children finish page 2.
20	38–70	**Check page 2** s-ws2ans	answers S3, S4, S5	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q3 a triangle, because the column has a circle and a square; Q4 No, the first column has two circles; Q5 3 circles.
21	38–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Two clues, how many shapes, what stays correct / Go further (page 4) | Four by Four, and bigger grids / Investigation (page 5) | Grid builder / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
22	70–76	**Talk and explain** s-talk	A CHILD SAYS: / I can fill every empty box by looking along the row. I never need the column. / Is the child right? Test it with your tiles. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Children test it with their tiles, using the grid from the sneak-peek: look at the bottom row, which has two empty boxes. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: not always. If a row has only one empty box, the row tells us. If a row has two empty boxes, the row cannot decide alone, so we must look down the column too. Repeat each child answer in your own words for the recording. The child who is wrong is a child in another class: no cast name is given the wrong answer.

23	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: circle. Puzzle 2: a square. The column has a triangle and a circle, and the row alone could not tell us.	Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone earlier answer. Puzzle 1: the row has a square and a triangle, so the circle is missing. Puzzle 2: the first box of the bottom row has a triangle and a circle above it, so it is a square; its row shows only a triangle, so the row could not decide alone. It was the same idea as the child in the talk question.
24	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, and the four options of E2. Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
25	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: triangle; option 2, square, triangle, circle; triangle. Do not read out any child result.
26	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
27	85–90	**Well done, inquirers!** s-end	Show a grown-up one grid you filled today. / NEXT LESSON: / Make a Table / Have ready: the strip page, cut out by a grown-up (14 strips).	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs the strip page, with the 14 strips cut out by a grown-up (no matches or toothpicks: paper strips only), and a pencil. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
28	optional	**Go further: three in a row** s-opt	Mei has a 3 by 3 grid with the digits 1, 2 and 3. / [picture: grid] / Every row and every column has 1, 2 and 3 once. Can you fill it?	Optional, only if you are on time. A 3 by 3 grid of the digits 1, 2 and 3 with 1, 2 and 3 on the long line from the top left to the bottom right. There is exactly one way to fill it: the top row must be 1, 3, 2 (a 2 in the top row would sit above the 2 in the middle), then 3, 2, 1 and 2, 1, 3. Take two answers: children say the first row, then check the rows and the columns together. Ask: how do you know the top row cannot be 1, 2, 3? Take the idea, not the speed.
29	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Row right, column right? Test it with the same row three times. / Make a puzzle | Cover three boxes. Ask a grown-up to find them. / Teach it back | Show a toy how to find a missing box. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your grid tiles. Pause or rewind any time.*	Gather your workbook, tiles, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Find the missing shape and the missing digit, and say which row has one of each. Pause if you need longer.*	Play along. Say each answer out loud before the video does.

8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Put one of each in every row, then in every column, with your tiles. Press play when you have tried.*	Pause. Fill mat 1 so every row and every column has one of each. Then try again with the same top row.
23:00	Mini-lesson	*Recording: keep watching and copy the examples with your tiles. Rewind if you miss a step.*	Watch and copy the examples with your tiles. Rewind any step you missed.
34:00	Brain break	*Recording: stand up and join in! Make the shapes with your body as I call them.*	Stand up. Make each shape with your arms as it is called.
38:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up whether the child is right, and test it with your tiles.*	Pause. Explain your answer to someone at home, and test it with your tiles.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one grid you filled.*	Show a grown-up your workbook and one grid you filled.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:33	1:45	ok
remember	6:00	4:53	5:46	ok
sneak	4:00	3:23	3:57	ok
explore	11:00	8:36	10:12	ok
mini	11:00	8:31	10:01	ok
break	4:00	3:22	4:00	ok
work	32:00	26:03	31:01	ok
talk	6:00	4:26	5:15	ok
exit	6:00	4:42	5:32	ok
look	3:00	2:23	2:48	ok
flex	5:00	0:44	0:50	ok
Total	90:00	68:36	81:08	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 5 questions; a middle child needs about 6.0 min for page 1 and 8.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.